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Simulation of LEO Satellite Integrated Autonomous FANET Network

CENG797 Ad Hoc Networks
2025 Fall
Term Project Report

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<https://github.com/hks1444/CENG-797-Ad-Hoc-Networks-Term-Project>

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1 Introduction

There are various studies on the use of FANET (Flying Ad Hoc Networks), LEO satellites (Low Earth Orbit), and LoRa technologies in IoT communication applications. However, no study integrates FANETs, LEO satellite systems, and LoRa technologies within an AI-supported network stack.

Such cross-integration would enable the development of an autonomous, energy-efficient, and low-latency communication infrastructure adaptable to dynamic topologies. This infrastructure would leverage artificial intelligence to autonomously manage processes such as end-to-end data transmission, routing, channel access, and resource management, optimizing network performance in real time. The approach provides an innovative, scalable, and resilient foundation for critical domains such as disaster management, agriculture, and environmental monitoring, offering significant opportunities for both academic and industrial applications.

The project aims to overcome the limitations of existing communication systems by implementing cross-integration of FANETs, LEO satellite systems, and LoRa technologies within an AI-supported network stack. An energy-efficient, and low-latency communication system that enables long-range connectivity for IoT devices, optimizes routing, channel access, and resource management with minimal human intervention, supports rapid and reliable big data transmission and in-network processing, provides a scalable and resilient infrastructure adaptable to changing environments, and ensures reliable operation across domains such as disaster management, agriculture, and environmental monitoring will be developed.

The hypothesis of the project is the following: An autonomous, energy-efficient, low-latency communication infrastructure adaptable to dynamic topologies that integrates FANETs, LEO satellite systems, and LoRa technologies within an AI-supported network stack can be created.

In conclusion, the study makes three primary contributions:

- It simulates a hybrid network architecture integrating FANETs, LEO satellite systems, and LoRa technologies.
- It designs an AI-supported network stack that optimizes decision-making by monitoring the overall network status in real time.
- It simulates a communication infrastructure applicable to domains such as disaster management, environmental monitoring, and agriculture.

Keywords: FANET, LEO satellites, LoRa, LoRaWAN, Reinforcement Learning, IoT

2 Background and Related Work

2.1 Background

The main difference between ad hoc networks and cellular networks is the existence of a fixed infrastructure. Ad hoc networks lack a fixed infrastructure and they are powered by a battery. This requires ad hoc networks to perform computation in a distributed manner. VANETs (Vehicular Ad Hoc Networks), MANETs (Mobile Ad Hoc Networks), and FANET (Flying Ad Hoc Networks) are the some of the main subcategories of ad hoc networks.

FANET facilitates connection between flying nodes like UAVs to support various applications

and tasks. Multiple UAVs can be deployed to fly over a targeted region and to perform surveillance, detection, and monitoring. The UAVs use sensors and cameras to capture real-time images, audios, and videos at remote and difficult-to-access areas and send the data to a base station immediately [1].

LEO (Low Earth Orbit) satellite systems form networks with satellites orbiting below 2,000 km. They offer low latency, high data rates and wide coverage, making them ideal for global connectivity and IoT applications. LoRaWAN (Long Range Wide Area Network), a low-power wide-area technology, enables long-range, energy-efficient communication between IoT devices. When integrated with LEO satellites, LoRa enhances connectivity in remote areas. However, LEO satellites provide limited coverage duration because they orbit the Earth rapidly, meaning any given region is only within their line of sight for a short period.

Combining FANET, LEO satellites, and LoRa supports adaptive, low-latency communication in dynamic topologies. An AI-driven network stack can further minimize human intervention, enabling autonomous, scalable, and efficient communication infrastructures.

2.2 Related Work

Recently, UAVs have been increasingly utilized in diverse applications and services, and numerous studies have proposed methods to enhance their practical deployment. A study [2] discusses livestock management research in which UAVs monitor farm animals and carry out tasks such as animal detection, counting, tracking, etc. However, the studies it reviews mostly rely on pre-established infrastructure to organize UAV connectivity and data collection. For example, this study [3] requires a local network built over long-distance WiFi access points, and another study [4] employs base station infrastructure and satellites. An additional study [5] focuses on the use of UAVs for smart farming and is based on existing infrastructure. The usage of FANET and LEO satellite systems can reduce costs by eliminating the need for fixed infrastructure, while also extending coverage and enabling the network to scale to more UAVs without manual configuration or explicit planning, thus enhancing system scalability.

In another article [6], the authors surveyed the use of unmanned aerial vehicles (UAV), altitude platforms (AP), and low-Earth orbit (LEO) satellites to deliver broadband services to remote and low-income areas. They concluded that UAVs are a promising solution due to their high maneuverability. A different study [7] proposed a location-aided delay tolerant routing protocol for UAV networks in post-disaster operations, leveraging location-aided forwarding in combination with a store-carry-forward (SCF) technique. However, these studies did not examine an integrated approach combining both LEO satellites and FANET.

FANET exhibit high mobility and dynamic topology, making their operation unpredictable. Optimal network designs must be adaptive, fast, scalable, energy-efficient, stable, and secure. Reinforcement Learning (RL) is well-suited for this purpose due to its self-learning and enables UAVs to make autonomous, context-aware decisions [8]. Researchers have proposed various reinforcement learning-based routing protocols and flight trajectory selection methods in FANET.

The following reinforcement-learning-based routing methods were reviewed in [9]. DQN-VR [10] introduces a DQN-based vertical clustering and routing for 5G FANET combining centralized and distributed control. QTAR [11] is a Q-learning-based topology-aware method for FANET using two-hop neighbor data to cut hops and avoid loops. TQNGPSR [12] extends greedy perimeter stateless routing with a Q-network that penalizes congested neighbors via

buffer queue length feedback. QMR [13] is multi-objective Q-learning optimization routing protocol for FANET. They attempt to reduce delay and energy. They proposed a method of adaptively adjusting Q-learning parameters and a new exploration and exploitation mechanism in Q-learning. QGeo [14] is a distributed Q-learning-based geographic protocol. The reward function is calculated based on the packet speed and the discount factor is adjusted according to the distance and speed of UAVs in the network. ICRA [15] increases the stability of the network topology and improves network lifetime by balancing energy consumption in the network via clustering whose utility weights (energy/centrality/mobility/link time) are tuned by centralized Q-learning. TARRAQ [16] adapts Q-learning using the neighbor change rate and neighbors' change inter-arrival time distribution to lower routing overhead during neighbor discovery. The learning rate and discount factor are computed from the residual link duration to minimize delay and energy consumption while improving packet delivery rate. PPMAC+RLSRP [17] is an adaptive communication protocol for FANETs consisting of a position-prediction-based directional MAC protocol that includes position prediction, communication control, and data transfer steps, and an RL-based self-learning routing protocol that updates routing policies based on UAV positions and defines a reward function using transmission delay. QFL-MOR [18] is a Q-learning-based fuzzy logic for multi-objective routing protocol, scoring links (transmission rate/energy status/flight status) and paths (hop count/successful packet delivery time) to refine routes iteratively.

The following reinforcement-learning-based UAV trajectory optimization approaches were reviewed in [8]. [19] is a Q-learning-based UAV flight planning algorithm that enhances video users' Quality of Experience (QoE). Multiple UAV base stations (UBSs) hover over user clusters, providing prefetched or on-demand video streams. The Q-learning agent selects flight paths considering cluster demand, recharging requirements, and flight time, following an ϵ -greedy policy. In [20], multiple UAVs do real-time sensing and sending using an algorithm that learns other agents' behavior as stationary distributions to select joint actions efficiently. Similarly, in [21], a decentralized DRL algorithm was used to navigate multiple UBSs to provide data services to a set of ground users. In [22], authors predicted the movement and future positions of the users and determined trajectories toward the users to deliver data with minimum power by using an echo state network (ESN) and multiagent Q-learning algorithm. In [23], UAVs are deployed in 3D to maximize user mean opinion score (MOS) and QoE. A three-step method clusters users with k-means, then applies Q-learning for 3D positioning and trajectory control. Extensions using deep or double Q-learning [24] [25] improved scalability and user satisfaction. Bayerlein et al. [26] optimized UBS trajectories to maximize the sum rate for multiple users while avoiding obstacles, using both table-based and neural network-based Q-learning. Similarly, Klaine et al. [27] applied Q-learning to control UBSs for emergency radio coverage in disaster areas, maximizing coverage and sum rate while mitigating interference and obstacles.

3 Main Contributions

You can introduce multiple sections for presenting your contribution. Provide content specific titles.

3.1 Your Contribution 1

Present your proposals, algorithms, techniques etc. here.

3.2 Your Contribution 2

Present your proposals, algorithms, techniques etc. here.

4 Results and Discussion

4.1 Methodology

Writing the methodology lies at the core of the paper, and fulfills one of the basic principles underlying the scientific method. Any scientific paper needs to be verifiable by other researchers, so that they can review the results by replicating the experiment and guaranteeing the validity. To assist this, you need to give a completely accurate description of the equipment and the techniques used for gathering the data [28].

Other scientists are not going to take your word for it, and they want to be able to evaluate whether your methodology is sound. In addition, it is useful for the reader to understand how you obtained your data, because it allows them to evaluate the quality of the results. For example, if you were trying to obtain data about shopping preferences, you will obtain different results from a multiple-choice questionnaire than from a series of open interviews. Writing methodology allows the reader to make their own decision about the validity of the data. If the research about shopping preferences were built upon a single case study, it would have little external validity, and the reader would treat the results with the contempt that they deserve [28].

Describe the materials and equipment used in the research. Explain how the samples were gathered, any randomization techniques and how the samples were prepared. Explain how the measurements were made and what calculations were performed upon the raw data. Describe the statistical techniques used upon the data [28].

4.2 Results

This is probably the most variable part of any research paper, and depends upon the results and aims of the experiment. For quantitative research, it is a presentation of the numerical results and data, whereas for qualitative research it should be a broader discussion of trends, without going into too much detail. For research generating a lot of results, then it is better to include tables or graphs of the analyzed data and leave the raw data in the appendix, so that a researcher can follow up and check your calculations. A commentary is essential to linking the results together, rather than displaying isolated and unconnected charts, figures and findings. It can be quite difficult to find a good balance between the results and the discussion section, because some findings, especially in a quantitative or descriptive experiment, will fall into a grey area. As long as you not repeat yourself too often, then there should be no major problem. It is best to try to find a middle course, where you give a general overview of the data and then expand upon it in the discussion - you should try to keep your own opinions and interpretations out of the results section, saving that for the discussion [28].

4.3 Discussion

This is where you elaborate upon your findings, and explain what you found, adding your own personal interpretations. Ideally, you should link the discussion back to the introduction, addressing each initial point individually. It is important to try to make sure that every

piece of information in your discussion is directly related to the thesis statement, or you risk clouding your findings. You can expand upon the topic in the conclusion - remembering the hourglass principle [28].

5 Conclusion

In general a short summarizing paragraph will do, and under no circumstances should the paragraph simply repeat material from the Abstract or Introduction. In some cases it's possible to now make the original claims more concrete, e.g., by referring to quantitative performance results [29].

The conclusion is where you build upon your discussion and try to refer your findings to other research and to the world at large. In a short research paper, it may be a paragraph or two, or practically non-existent. In a dissertation, it may well be the most important part of the entire paper - not only does it describe the results and discussion in detail, it emphasizes the importance of the results in the field, and ties it in with the previous research. Some research papers require a recommendations section, postulating that further directions of the research, as well as highlighting how any flaws affected the results. In this case, you should suggest any improvements that could be made to the research design [28].

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Appendix A About Appendices

Appendices should contain detailed proofs and algorithms only. Appendices can be crucial for overlength papers, but are still useful otherwise. Think of appendices as random-access substantiation of underlying gory details. As a rule of thumb: (1) Appendices should not contain any material necessary for understanding the contributions of the paper. (2) Appendices should contain all material that most readers would not be interested in [29].

Appendix B Assessment

Your report will be assessed based on the following list of criteria.

Appendix B.1 Style

[15 points] The report states title, author names, affiliations and date. The format follows this style?

1. Structure and Organization: Does the organization of the paper enhance understanding of the material? Is the flow logical with appropriate transitions between sections?
2. Technical Exposition: Is the technical material presented clearly and logically? Is the material presented at the appropriate level of detail?
3. Clarity: Is the writing clear, unambiguous and direct? Is there excessive use of jargon, acronyms or undefined terms?
4. Style: Does the writing adhere to conventional rules of grammar and style? Are the references sufficient and appropriate?
5. Length: Is the length of the paper appropriate to the technical content?
6. Illustrations: Do the figures and tables enhance understanding of the text? Are they well explained? Are they of appropriate number, format and size?

Appendix B.2 Abstract

[10 points] Does the abstract summarize the report? These are the basic components of an abstract in any discipline:

1. Motivation/problem statement: Why do we care about the problem? What practical, scientific, theoretical or artistic gap is your research filling?
2. Methods/procedure/approach: What did you actually do to get your results? (e.g. analyzed 3 novels, completed a series of 5 oil paintings, interviewed 17 students)
3. Results/findings/product: As a result of completing the above procedure, what did you learn/invent/create?
4. Conclusion/implications: What are the larger implications of your findings, especially for the problem/gap identified?

Appendix B.3 The Problem

[15 points] The problem section must be specific. The title of the section must indicate your problem. Do not use generic titles.

1. Is the problem clearly stated?
2. Is the problem practically important?
3. What is the purpose of the study?
4. What is the hypothesis?
5. Are the key terms defined?

Appendix B.4 Background and Related Work

[15 points] Does the report present the background and related work in separate sections.

1. Are the cited sources pertinent to the study?

2. Is the review too broad or too narrow?
3. Are the references recent?
4. Is there any evidence of bias?

Appendix B.5 Design

[15 points] Does the report present the design of the study.

1. What research methodology was used?
2. Was it a replica study or an original study?
3. What measurement tools were used?
4. How were the procedures structured?
5. Was a pilot study conducted?
6. What are the variables?
7. How was sampling performed?

Appendix B.6 Analysis

[15 points] Does the report present the analysis?

1. How was data analyzed?
2. Was data qualitative or quantitative?
3. Did findings support the hypothesis and purpose?
4. Were weaknesses and problems discussed?

Appendix B.7 Conclusion and Future Work

[15 points] Does the report state the conclusion and future work clearly?

1. Are the conclusions of the study related to the original purpose?
2. Were the implications discussed?
3. Whom the results and conclusions will effect?
4. What recommendations were made at the conclusion?